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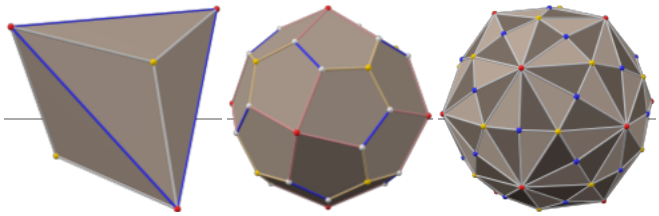
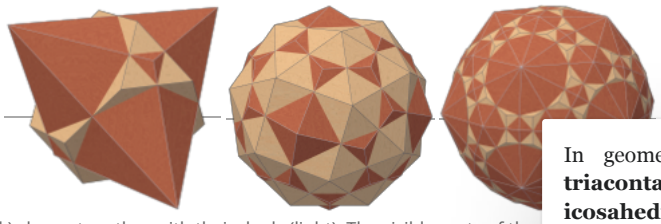
Catalan solid

In mathematics, a **Catalan solid**, or **Archimedean dual**, is a polyhedron that is dual to an Archimedean solid. There are 13 Catalan solids. They are named after the Belgian mathematician Eugène Catalan, who first described them in 1865.



Triakis tetrahedron, pentagonal icositetrahedron and disdyakis triacontahedron.

The Catalan solids are all convex. They are face-transitive but not vertex-transitive. This is because the dual Archimedean solids are vertex-transitive and not face-transitive. Note that unlike Platonic solids and Archimedean solids, the faces of Catalan solids are *not* regular polygons. However, the vertex figures of Catalan solids are regular, and they have constant dihedral angles. Being face-transitive, Catalan solids are isohedra.



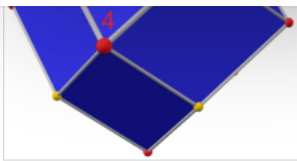
The solids above (dark) shown together with their duals (light). The visible parts of the

Additionally, two of the Catalan solids are edge-transitive: the rhombic dodecahedron and the rhombic triacontahedron. These are the duals of the two quasi-regular Archimedean solids.

Just as prisms and antiprisms are generally not considered Archimedean solids, bipyramids and trapezohedra are generally not considered Catalan solids, despite being face-transitive.

Two of the Catalan solids are chiral: the pentagonal icositetrahedron and the pentagonal hexecontahedron, dual to the chiral snub cube and snub dodecahedron. These each come in two enantiomorphs. Not counting the enantiomorphs, bipyramids, and trapezohedra, there are a total of 13 Catalan solids.

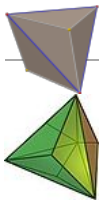
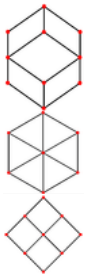
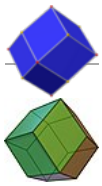
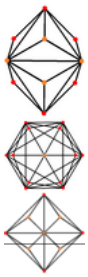
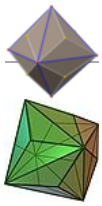
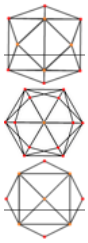
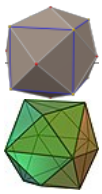
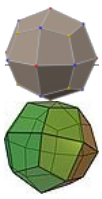
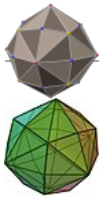
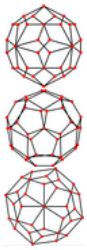
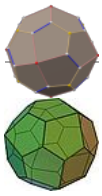
In geometry, a **disdyakis triacontahedron**, **hexakis icosahedron**, **decakis dodecahedron** or **kisrhombic triacontahedron** is a Catalan solid with 120 faces and the dual to

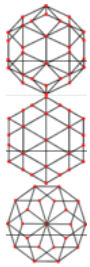
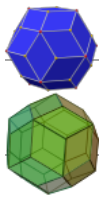
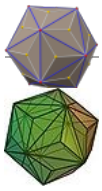
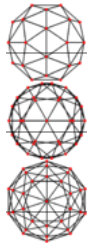
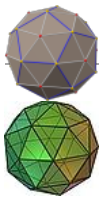
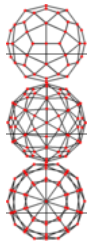
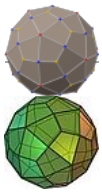
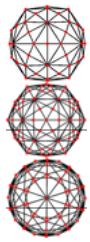
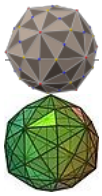
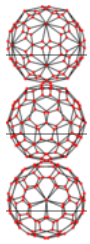
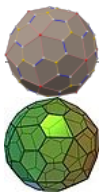


A rhombic dodecahedron with its face configuration.

Eleven of the 13 Catalan solids have the Rupert property: a copy of the solid, of the same or larger shape, can be passed through a hole in the solid. [1]

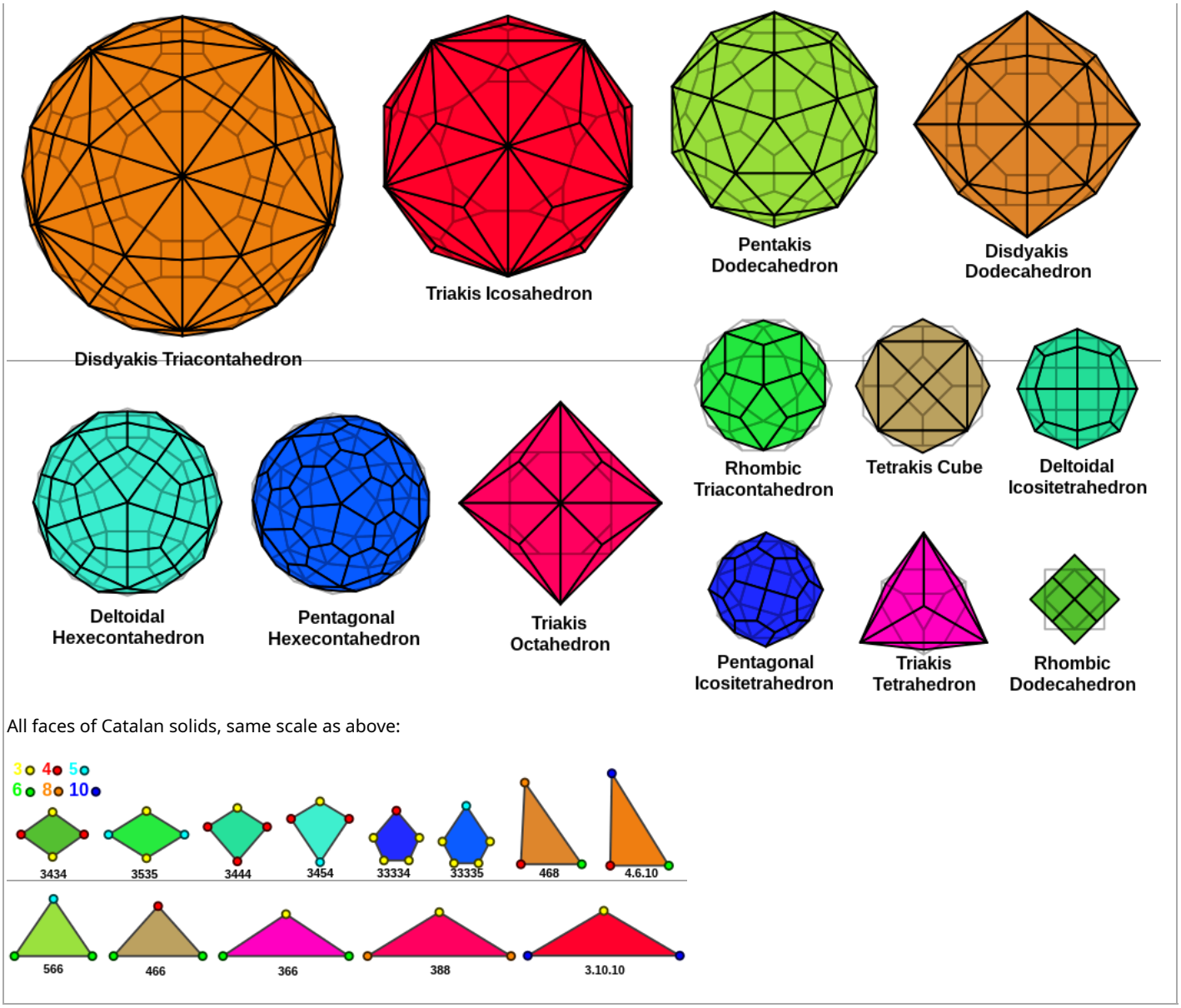
List of Catalan solids and their duals

Conway name	Archimedean dual	Face polygon	Orthogonal wireframes	Pictures	Face angles (°)	Dihedral angle (°)	Midradius[2]	Faces	Edges	Vert	Sy
triakis tetrahedron "kT"	truncated tetrahedron	Isosceles V3.6.6			112.885 33.557 33.557	129.521	1.0607	12	18	8	
rhombic dodecahedron "jC"	cuboctahedron	Rhombus V3.4.3.4			70.529 109.471 70.529 109.471	120	0.8660	12	24	14	
triakis octahedron "kO"	truncated cube	Isosceles V3.8.8			117.201 31.400 31.400	147.350	1.7071	24	36	14	
tetrakis hexahedron "kC"	truncated octahedron	Isosceles V4.6.6			83.621 48.190 48.190	143.130	1.5000	24	36	14	
deltoidal icositetrahedron "oC"	rhombicuboctahedron	Kite V3.4.4.4			81.579 81.579 81.579 115.263	138.118	1.3066	24	48	26	
disdyakis dodecahedron "mC"	truncated cuboctahedron	Scalene V4.6.8			87.202 55.025 37.773	155.082	2.2630	48	72	26	
pentagonal icositetrahedron "gC"	snub cube	Pentagon V3.3.3.3.4			114.812 114.812 114.812 114.812 80.752	136.309	1.2472	24	60	38	

Conway name	Archimedean dual	Face polygon	Orthogonal wireframes	Pictures	Face angles (°)	Dihedral angle (°)	Midradius[2]	Faces	Edges	Vert	Sy
rhombic triacontahedron "jD"	icosidodecahedron	Rhombus V3.5.3.5			63.435 116.565 63.435 116.565	144	1.5388	30	60	32	
triakis icosahedron "kI"	truncated dodecahedron	Isosceles V3.10.10			119.039 30.480 30.480	160.613	2.9271	60	90	32	
pentakis dodecahedron "kD"	truncated icosahedron	Isosceles V5.6.6			68.619 55.691 55.691	156.719	2.4271	60	90	32	
deltoidal hexecontahedron "oD"	rhombicosidodecahedron	Kite V3.4.5.4			86.974 67.783 86.974 118.269	154.121	2.1763	60	120	62	
disdyakis triacontahedron "mD"	truncated icosidodecahedron	Scalene V4.6.10			88.992 58.238 32.770	164.888	3.7694	120	180	62	
pentagonal hexecontahedron "gD"	snub dodecahedron	Pentagon V3.3.3.3.5			118.137 118.137 118.137 118.137 67.454	153.179	2.0971	60	150	92	

ordered by size [show]

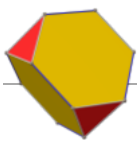
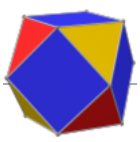
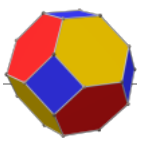
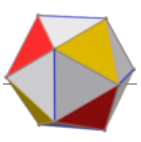
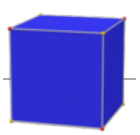
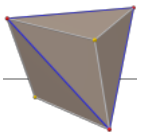
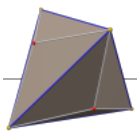
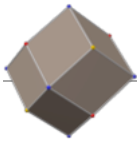
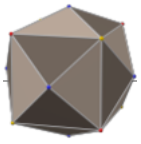
The Catalan solids' Archimedean duals, all shown with the same edge length. Sorted by midradius in descending order:



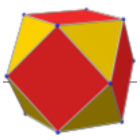
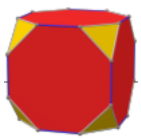
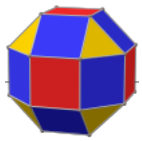
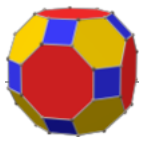
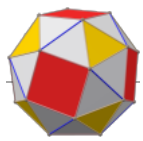
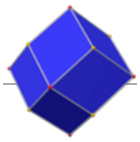
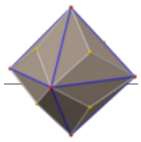
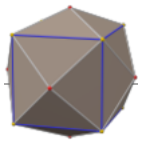
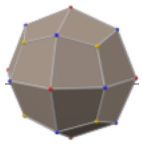
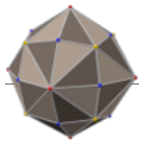
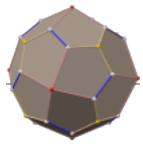
Symmetry

The Catalan solids, along with their dual Archimedean solids, can be grouped in those with tetrahedral, octahedral and icosahedral symmetry. For both octahedral and icosahedral symmetry there are six forms. The only Catalan solid with genuine tetrahedral symmetry is the triakis tetrahedron (dual of the truncated tetrahedron). The rhombic dodecahedron and tetrakis hexahedron have octahedral symmetry, but they can be colored to have only tetrahedral symmetry. Rectification and snub also exist with tetrahedral symmetry, but they are Platonic instead of Archimedean, so their duals are Platonic instead of Catalan. (They are shown with brown background in the table below.)

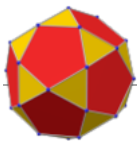
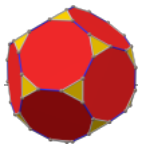
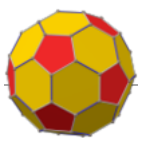
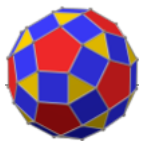
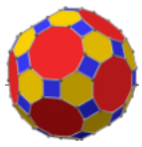
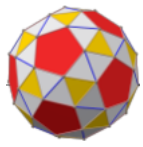
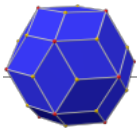
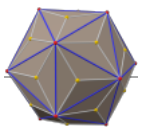
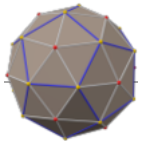
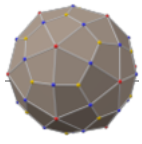
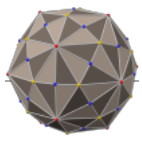
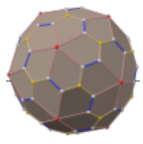
Tetrahedral symmetry

Archimedean (Platonic)						
Catalan (Platonic)						

Octahedral symmetry

Archimedean						
Catalan						

Icosahedral symmetry

Archimedean						
Catalan						

Geometry

All dihedral angles of a Catalan solid are equal. Denoting their value by θ , and denoting the face angle at the vertices where p faces meet by α_p, we have

$$\sin(\theta/2) = \cos(\pi/p)/\cos(\alpha_p/2).$$

This can be used to compute θ and α_p, α_q, ... , from p, q ... only.

Triangular faces

Of the 13 Catalan solids, 7 have triangular faces. These are of the form Vp.q.r, where p, q and r take their values among 3, 4, 5, 6, 8 and 10. The angles α_p, α_q and α_r can be computed in the following way. Put $a = 4\cos^2(\pi/p)$, $b = 4\cos^2(\pi/q)$, $c = 4\cos^2(\pi/r)$ and put

$$S = -a^2 - b^2 - c^2 + 2ab + 2bc + 2ca.$$

Then

$$\cos(\alpha_p) = \frac{S}{2bc} - 1,$$
$$\sin(\alpha_p/2) = \frac{-a+b+c}{2\sqrt{bc}}.$$

For α_q and α_r the expressions are similar of course. The dihedral angle θ can be computed from

$$\cos(\theta) = 1 - 2abc/S.$$

Applying this, for example, to the disdyakis triacontahedron ($p = 4$, $q = 6$ and $r = 10$, hence $a = 2$, $b = 3$ and $c = \phi + 2$, where ϕ is the golden ratio) gives $\cos(\alpha_4) = \dfrac{2-\phi}{6(2+\phi)} = \dfrac{7-4\phi}{30}$ and $\cos(\theta) = \dfrac{-10-7\phi}{14+5\phi} = \dfrac{-48\phi-155}{241}$.

Quadrilateral faces

Of the 13 Catalan solids, 4 have quadrilateral faces. These are of the form Vp.q.p.r, where p, q and r take their values among 3, 4, and 5. The angle α_p can be computed by the following formula:

$$\cos(\alpha_p) = \frac{2\cos^2(\pi/p) - \cos^2(\pi/q) - \cos^2(\pi/r)}{2\cos^2(\pi/p) + 2\cos(\pi/q)\cos(\pi/r)}.$$

From this, α_q, α_r and the dihedral angle can be easily computed. Alternatively, put $a = 4\cos^2(\pi/p)$, $b = 4\cos^2(\pi/q)$, $c = 4\cos^2(\pi/p) + 4\cos(\pi/q)\cos(\pi/r)$. Then α_p and α_q can be found by applying the formulas for the triangular case. The angle α_r can be computed similarly of course. The faces are kites, or, if $q = r$, rhombi. Applying this, for example, to the deltoidal icositetrahedron ($p = 4$, $q = 3$ and $r = 4$), we get $\cos(\alpha_4) = \dfrac{1}{2} - \dfrac{1}{4}\sqrt{2}$.

Pentagonal faces

Of the 13 Catalan solids, 2 have pentagonal faces. These are of the form Vp.p.p.p.q, where p=3, and q=4 or 5. The angle $\alpha_p$can be computed by solving a degree three equation:

$$8\cos^2(\pi/p)\cos^3(\alpha_p) - 8\cos^2(\pi/p)\cos^2(\alpha_p) + \cos^2(\pi/q) = 0.$$

Metric properties

For a Catalan solid $\mathbf{C}$ let $\mathbf{A}$ be the dual with respect to the midsphere of $\mathbf{C}$. Then $\mathbf{A}$ is an Archimedean solid with the same midsphere. Denote the length of the edges of $\mathbf{A}$ by l. Let r be the inradius of the faces of $\mathbf{C}$, r_m the midradius of $\mathbf{C}$ and $\mathbf{A}$, r_i the inradius of $\mathbf{C}$, and r_c the circumradius of $\mathbf{A}$. Then these quantities can be expressed in l and the dihedral angle θ as follows:

$$r^2 = \frac{l^2}{8}(1-\cos\theta),$$

$$r_m^2 = \frac{l^2}{4}\frac{1-\cos\theta}{1+\cos\theta},$$

$$r_i^2 = \frac{l^2}{8}\frac{(1-\cos\theta)^2}{1+\cos\theta},$$

$$r_c^2 = \frac{l^2}{2}\frac{1}{1+\cos\theta}.$$

These quantities are related by $r_m^2 = r_i^2 + r^2$, $r_c^2 = r_m^2 + l^2/4$ and $r_i r_c = r_m^2$.

As an example, let $\mathbf{A}$ be a cuboctahedron with edge length $l = 1$. Then $\mathbf{C}$ is a rhombic dodecahedron. Applying the formula for quadrilateral faces with $p = 4$ and $q = r = 3$ gives $\cos\theta = -1/2$, hence $r_i = 3/4$, $r_m = \frac{1}{2}\sqrt{3}$, $r_c = 1$, $r = \frac{1}{4}\sqrt{3}$.

All vertices of $\mathbf{C}$ of type p lie on a sphere with radius $r_{c,p}$ given by

$$r_{c,p}^2 = r_i^2 + \frac{2r^2}{1-\cos\alpha_p},$$

and similarly for $q, r, \ldots$.

Dually, there is a sphere which touches all faces of $\mathbf{A}$ which are regular p-gons (and similarly for $q, r, \ldots$) in their center. The radius $r_{i,p}$ of this sphere is given by

$$r_{i,p}^2 = r_m^2 - \frac{l^2}{4}\cot^2(\pi/p).$$

These two radii are related by $r_{i,p}r_{c,p} = r_m^2$. Continuing the above example: $\cos\alpha_3 = -1/3$ and $\cos\alpha_4 = 1/3$, which gives $r_{c,3} = \frac{3}{8}\sqrt{6}$, $r_{c,4} = \frac{3}{4}\sqrt{2}$, $r_{i,3} = \frac{1}{3}\sqrt{6}$ and $r_{i,4} = \frac{1}{2}\sqrt{2}$.

If P is a vertex of $\mathbf{C}$ of type p, e an edge of $\mathbf{C}$ starting at P, and P' the point where the edge e touches the midsphere of $\mathbf{C}$, denote the distance PP' by l_p. Then the edges of $\mathbf{C}$ joining vertices of type p and type q have length $l_{p,q} = l_p + l_q$. These quantities can be computed by

$$l_p = \frac{l}{2}\frac{\cos(\pi/p)}{\sin(\alpha_p/2)},$$

and similarly for $q, r, \ldots$. Continuing the above example: $\sin(\alpha_3/2) = \frac{1}{3}\sqrt{6}$, $\sin(\alpha_4/2) = \frac{1}{3}\sqrt{3}$, $l_3 = \frac{1}{8}\sqrt{6}$, $l_4 = \frac{1}{4}\sqrt{6}$, so the edges of the rhombic dodecahedron have length $l_{3,4} = \frac{3}{8}\sqrt{6}$.

The dihedral angles $\alpha_{p,q}$between p-gonal and q-gonal faces of $\mathbf{A}$ satisfy

$$\cos\alpha_{p,q} = \frac{l^2}{4}\frac{\cot(\pi/p)\cot(\pi/q)}{r_m^2} - \frac{r_{i,p}r_{i,q}}{r_m^2} = \frac{l_p l_q - r_m^2}{r_{c,p}r_{c,q}}.$$

Finishing the rhombic dodecahedron example, the dihedral angle $\alpha_{3,4}$ of the cuboctahedron is given by $\cos\alpha_{3,4} = -\frac{1}{3}\sqrt{3}$.

Construction

The face of any Catalan polyhedron may be obtained from the vertex figure of the dual Archimedean solid using the Dorman Luke construction.[3]

Application to other solids

All of the formulae of this section apply to the Platonic solids, and bipyramids and trapezohedra with equal dihedral angles as well, because they can be derived from the constant dihedral angle property only. For the pentagonal trapezohedron, for example, with faces V3.3.5.3, we get $\cos(\alpha_3) = \frac{1}{4} - \frac{1}{4}\sqrt{5}$, or $\alpha_3 = 108°$. This is not surprising: it is possible to cut off both apexes in such a way as to obtain a regular dodecahedron.

See also

- List of uniform tilings Shows dual uniform polygonal tilings similar to the Catalan solids
- Conway polyhedron notation A notational construction process
- Archimedean solid
- Johnson solid

Notes

1. Fredriksson, Albin (2024), "Optimizing for the Rupert property", *The American Mathematical Monthly*, **131** (3): 255–261, arXiv:2210.00601 (https://arxiv.org/abs/2210.00601), doi:10.1080/00029890.2023.2285200 (https://doi.org/10.1080%2F00029890.2023.2285200)
2. Weisstein, Eric W. "Archimedean Solid" (https://mathworld.wolfram.com/). *mathworld.wolfram.com*. Retrieved 2022-07-02.
3. Cundy & Rollett (1961), p. 117; Wenninger (1983), p. 30.

References

- Eugène Catalan *Mémoire sur la Théorie des Polyèdres.* J. l'École Polytechnique (Paris) 41, 1-71, 1865.
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- Gailiunas, P.; Sharp, J. (2005), "Duality of polyhedra", *International Journal of Mathematical Education in Science and Technology*, **36** (6): 617–642, doi:10.1080/00207390500064049 (https://doi.org/10.1080%2F00207390500064049), S2CID 120818796 (https://api.semanticscholar.org/CorpusID:120818796).
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- Wenninger, Magnus (1983), *Dual Models*, Cambridge University Press, ISBN 978-0-521-54325-5, MR 0730208 (https://mathscinet.ams.org/mathscinet-getitem?mr=0730208) (The thirteen semiregular convex polyhedra and their duals)
- Williams, Robert (1979). *The Geometrical Foundation of Natural Structure: A Source Book of Design*. Dover Publications, Inc. ISBN 0-486-23729-X. (Section 3-9)
- Anthony Pugh (1976). *Polyhedra: A visual approach*. California: University of California Press Berkeley. ISBN 0-520-03056-7. Chapter 4: Duals of the Archimedean polyhedra, prisma and antiprisms

External links

- Weisstein, Eric W. "Catalan Solids" (https://mathworld.wolfram.com/CatalanSolid.html). *MathWorld*.
- Weisstein, Eric W. "Isohedron" (https://mathworld.wolfram.com/Isohedron.html). *MathWorld*.
- Catalan Solids (http://dmccooey.com/polyhedra/Catalan.html) – at Visual Polyhedra
- Archimedean duals (http://www.georgehart.com/virtual-polyhedra/archimedean-duals-info.html) – at Virtual Reality Polyhedra
- Interactive Catalan Solid (https://web.archive.org/web/20051201035511/http://ibiblio.org/e-notes/3Dapp/Catalan.htm) in Java
- Download link for Catalan's original 1865 publication (https://orbi.uliege.be/handle/2268/194785) – with beautiful figures, PDF format

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